Errata in the Textbook — Casella & Berger, Statistical Inference (2e)

Contents

Chapter 1 — Probability Theory **1**

Chapter 2 — Transformations and Expectations **2**

Chapter 3 — Common Families of Distributions **2**

Chapter 4 — Multiple Random Variables **2**

Chapter 5 — Properties of a Random Sample **2**

Chapter 6 — Principles of Data Reduction **2**

Chapter 7 — Point Estimation **2**

Chapter 8 — Hypothesis Testing **3**

Chapter 9 — Interval Estimation **3**

Chapter 10 — Asymptotic Evaluations **3**

Chapter 11 — Analysis of Variance and Regression **3**

Chapter 12 — Regression Models **4**

Companion to "Casella & Berger, Statistical Inference (2nd edition) — Exercises with Solutions, All Chapters."

Errors in the printed text of Casella & Berger, *Statistical Inference*, 2nd edition, 2024 CRC Press / Chapman & Hall reprint (hardback ISBN 978-1-032-59303-6; paperback ISBN 978-1-032-59794-2), found while working the exercises: false statements, wrong formulas, wrong numerical values, and missing hypotheses. Each entry names the defect and gives the correction; the full counterexample or proof is in the solution to that exercise. 38 entries across 11 of the 12 chapters.

Chapter 1 — Probability Theory

1.29(d) — false identity. The weighted sum equals m^k by the multinomial theorem, not $\binom{k+m-1}{k}$ (e.g. $m = k = 2$ gives $1+2+1 = 4 \neq 3$). Stars and bars counts the *summands*; the authors' errata change the exercise to ask for that count.

1.31 — wrong value. The average of $\{2, 4, 9, 12\}$ is $27/4$, not $29/4$ as printed.

1.42(e) — wrong cross-reference (minor). The proof needs the alternating-binomial identity from part (d), not (a)–(c).

1.43(b) — false in both directions. Neither the printed inequality nor its reverse holds in general for the unnormalized bounds. What is true: the normalized averages $P_k/\binom{n}{k}$ decrease in k.

Chapter 2 — Transformations and Expectations

2.29(c) — wrong factor. The printed $a(y+a)$ cannot be a pmf; the beta-binomial pmf needs the factor $\frac{a}{y+a}$ (confirmed by the authors' errata).

2.40 — missing hypothesis. The identity requires integer $0 \le x < n$; at $x = n$ the right side is the indeterminate form $0 \cdot \infty$ (the $(n-x)$ factor vanishes but $\int_0^{1-p} t^{-1}(1-t)^n\,dt$ diverges for $p < 1$), not 0.

Chapter 3 — Common Families of Distributions

No errata found.

Chapter 4 — Multiple Random Variables

4.8 — missing Jacobian. With a continuous prior density, the two envelope contributions at observed x are $\pi(x)/2$ and $\pi(x/2)/4$: the doubled-envelope term needs the scale change of variables. Switching is favorable iff $\pi(x/2) < 4\pi(x)$. The text's factor-2 rule holds only for a discrete prior mass function.

4.37(a) — false as stated. The uniform-mixing moment argument does not prove the claim for fixed finite n; an explicit mixture of shifted beta-binomial laws is a counterexample.

Chapter 5 — Properties of a Random Sample

5.38(b) — missing restriction. "$P(S_n > a) \le c^n$ for all n" is false for $a < 0$ (e.g. $X \equiv -1$, $a = -2$ gives $P(S_n > -2) = 1$); it holds for $a \ge 0$, or only eventually for $a < 0$. Parts (c)–(d) use $a = 0$ and are unaffected.

5.56 — wrong density and cdf. $1/(1+y^2)$ is not a pdf ($\int = \pi$). The Cauchy density is $\frac{1}{\pi(1+y^2)}$ with cdf $\frac{1}{2} + \frac{1}{\pi}\tan^{-1}(y)$.

5.61(b) — false as stated. An equal-scale integer-shape gamma proposal cannot dominate the target; a finite rejection envelope needs a larger proposal scale (the authors' errata specify $b+1$).

Chapter 6 — Principles of Data Reduction

6.27(b) — misprinted statistic. The chi-square pivot needs $n/\bar{X}$, not $1/\bar{X}$, in the denominator.

Chapter 7 — Point Estimation

7.45(d) — wrong formula. The minimizing factor uses $K-3$ where $K = E(X-\mu)^4/\sigma^4$ is the standardized kurtosis, not the raw fourth moment minus 1.

7.51 — wrong cross-reference. "As in Exercise 7.46" should be "as in Exercise 7.50": 7.46 treats the uniform$(\theta, 2\theta)$ family and defines no cS; the $n(\theta, a\theta^2)$ notation with cS is from 7.50 (which part (b) already cites as 7.50b).

Chapter 8 — Hypothesis Testing

8.43 — wrong pivots and optimizer. As printed, the t statistic is not Student t; the F claim is wrong in both degrees-of-freedom order and scaling; the ρ^2 optimizer uses variances where standard deviations belong.

Chapter 9 — Interval Estimation

9.20(b) — reversed inequalities. Both inequalities must be reversed; coverage is at least $1-\alpha$, with equality not guaranteed for discrete laws.

9.49(b) — false "more powerful" claim. The power comparison is invalid as a uniform claim: because $(\alpha-p)/(1-p) < \alpha$, the cutoff $z_{(\alpha-p)/(1-p)}$ exceeds z_α, and the powers cross — test (a) wins for small shifts, test (b) only for large ones (e.g. $\alpha = .1$, $p = .05$: at $\delta = 1$, .376 vs .304; at $\delta = 5$, .961 vs 1.000).

9.45(b) — wrong cutoff. The UMA interval needs the lower-α (upper-$(1-\alpha)$) chi-square cutoff. In (d), the minimum-based interval has coverage 0.2993803913, not exactly 0.3.

9.30 — misleading aside (minor). The decay argument holds for every finite $b > 0$, including $b = 1/n$; it fixes n, a, b while the observed sum grows.

Chapter 10 — Asymptotic Evaluations

10.11(b) — false claim. With both gamma parameters unknown, $\hat{\mu}_{\text{MLE}} = \bar{X}$ exactly, so the ARE comparison is 1, not $\alpha\psi'(\alpha)$.

10.11(c) — false. The known-mean method-of-moments and maximum-likelihood scale estimators are different statistics (e.g. 0.156 versus 1.867 on the book's data).

10.16(a) — wrong value. The second-order term is 0.0002140103, not 0.00007.

10.16(b) — wrong value. The exact binomial variance is 0.0002710672072, not 0.00529.

10.19(b) — sign error. The term before $\frac{1-\rho^n}{1-\rho}$ should be $-$, not $+$.

10.29(b)(ii) — wrong function. For the stated logistic density the efficient score is $\tanh(x/2)$, not $\tanh(x)$.

(10.3.8) — wrong denominator. It must estimate the square of the *central* probability, not the outside probability.

(10.2.4) — sign slip. The Taylor ratio in the displayed expansion has a sign error (harmless for the subsequent asymptotics).

10.33 — inverted statistic. The pivot must multiply by observed information, $\sqrt{J_n(\hat{\theta})}(\hat{\theta}-\theta_0)$; the printed division is $O_p(n^{-1})$ and converges to zero. (The same inversion appears in the proof of Theorem 10.3.1.)

10.17 — data typo. The third pair prints as $(5.58, 2.81)$; the canonical Efron (1982) law-school dataset has $(558, 2.81)$. With the printed value the correlation drops to $r = 0.493$ (vs 0.776 with the intended value).

10.45 — false coverage claim. The parenthetical asserts the continuity-corrected interval maintains coverage above $1-\alpha$ for all p, but at $n = 20$ it dips to $.9490 < .95$.

Chapter 11 — Analysis of Variance and Regression

11.29(c) — sign error. $\text{Cov}(Y_i, \hat{\alpha})$ needs a minus sign before $\frac{(x_i-\bar{x})\bar{x}}{S_{xx}}$ (the following exercise prints it correctly).

11.40(a) — "max" should be "sup". The supremum is not attained, so "maximum" is incorrect.

Chapter 12 — Regression Models

12.3(b) — false. Squared distance violates the triangle inequality; $\sqrt{D_\lambda}$ is the metric.

12.27(a) — false as stated. The CLT needs the true parameter and the weighted sign score; the plug-in version fails (median counterexample).

12.27(b) — invalid proof step. The empirical LAD score is piecewise constant and cannot be differentiated termwise; the expected-score derivative is $-2f(0)q_n$.

12.26(a)(ii) — non sequitur. Bounded x_i does not imply convergence of the stated sequence; a further argument is needed.

12.31(c) — self-reference. Prints "Do (c) B times"; should be "Do (b) B times" (repeat the resampling step — (c) cannot refer to itself).

12.30(e) — naming. "Laplace" and "double exponential" are the same law, not two error distributions to compare.